

Abstract Algebra - Questions

- when is A invertible? Does it have to be square?
- what is a Group and an abelian group? What is the Symmetric group?
- Definitions for bijection, homomorphism, isomorphism and automorphism.
- How many bijections are there for n elements? ~~injection, surjection, and bijection of a set~~
- Definition of a subgroup and a cyclic one. Is a cyclic subgroup abelian?
- What is $\text{Aut}(G)$, $\text{Im}(f)$, $\text{Ker}(f)$?
- Definition of a normal subgroup and of a conjugation of n by g .
- How to show that $H \triangleleft G$?
- What is the center of G , $Z(G)$? When is the following true: $G = Z(G)$?
- Show that kernels are always normal subgroups.
- What is an equivalence relation on a set S and the equivalent classes $[S]$?
- What is a coset and what happens when the set is finite?
- for $H \leq G$ what is the index of H , $[G:H]$?
- What is the Lagrange theorem? (for $H \leq G$ and G finite)
- What happens if G is finite and $|G| = p$ prime?
- When is a group simple?
- When is a homomorphism injective? (think of the kernel)
- How to prove a collection of subsets is a partition of a set?
- What is $\bar{a}$ and what is $\mathbb{Z}/n\mathbb{Z}$, is it cyclic and why?
- definition of $\gcd(u, m)$, when are u and m relatively prime?
- What is the form of all of subgroups of $\mathbb{Z}$, prove it.
- What is the quotient group G/H ? Is every normal subgroup the kernel of some homomorphism?
- Explain the logic behind the kernel.
- Explain the isomorphism theorem. What does $G \cong H$ mean?
- What is a semigroup, a ~~group~~^{set} S ?
- What are the n -th roots of unity and what is the geometric shape of them?
- What is a vector space over a field F ? (2 main properties)
- Definition of a field of a set F . $((F, +), (F^*, \cdot))$ and distributive law, is $\mathbb{Z}$ a field?
- Why is $F = \{0, 1\}$ true? When is $\mathbb{Z}/n\mathbb{Z}$ a field?
- Definition of a linear transformation. What's a linear combination of vector $\vec{v}$?
- What's the subspace spanned by a set? what's $V = \text{Span } S$ of S ?

- When is a set a basis of a vector space? What's $\dim(V)$? What is it equal to?
- What directions does $\forall w$ keep track of?
- Explain the First Isomorphism Theorem and the logic of $\ker(T)$?
- What's rank and nullity of T , linear transformation? What's an endomorphism?
- What's ~~an~~ an invariant subspace and its invariant complement and the matrix $A = \dots$?
- What's the set of eigenvalues equal to given the linear operator $T - \lambda I$?
- Explain the Cayley-Hamilton theorem.
- What is $O_n(F)$ and $SO_n(F)$?
- Explain the logic of what $\langle v, w \rangle$ shows and write the Cauchy-Schwarz inequality.
- What is the index of a subgroup H in G ?
- What is a permutation matrix?
- What do transformations in $O_n(2)$ look like when $\det(A) = +1$ and when $\det(A) = -1$?
- What is the group G of rigid motions? what is G equal to?
- What's the point p that finite subgroups Γ of $G = \mathbb{R}^2 \cdot O(2)$ fix? (Proof)
- Given a discrete subgroup $\Gamma \subset G = \mathbb{R}^2 \cdot O(2)$ what is L and $\bar{\Gamma}$? What does $\bar{\Gamma}$ preserve?
- Can a rotation/reflection turn a lattice translation into a non-translation?
- What is C_n and D_n ? What's the relationship between an eigenvalue λ and an angle θ ?
- What is D_4 for rotation D_{2n} , and its relation with the Klein 4-group?
- What is the stabilizer and orbit of s ? What's G_s under conjugation?
- Explain ~~the~~ Cayley's theorem, about a group G and subgroups of S_n .
- What is a transitive action and its relation with O_s ?
- What happens to O_s when G is abelian?
- What's A_n ?
- Do conjugate elements have the same order?
- What is the normalizer of H , $N(H)$? What if $N(H) = G$ for $H \subset G$?
- Are ~~the~~ subgroups of index 2 normal?
- What is ring, subring, endomorphism ring, ideal and unit?
- What is $\mathbb{Z}[a]$? What's $\mathbb{Z}[i]$?
- What does it mean to be algebraic and what transcendental?
- How many ideals does a ~~ring~~ ^{field} have? When is $R = \mathbb{Z}/n\mathbb{Z}$ a field?
- What's the product ideal IJ ?

- If $p \equiv 1 \pmod{4}$, then what do we know about $\mathbb{Z}[i]$'s ideals?
- Explain how the order of G is the ~~order~~^{sum} of the orders of conjugacy classes.
- What is $\{G \curvearrowright S \text{ orbits}\}$ equivalent to?
- How to construct the quotient field? $\text{Frac}(R) = S/u$ with $S = R \setminus \{0\}$.
- What is a Euclidean domain?
- State Gauss's lemma and prove it.
- What rings do algebraic integers in $\mathbb{Q}(\sqrt{d})$ form? All the cases for d square-free.
- Talk about the golden mean.
- What is a maximal ideal and what is the product of ideal $I \cdot J$?
- If P prime ideal, then what do we know about R/P ?
- What is a Dedekind domain and the class group $\text{Cl}(R)$? What's a fractional ideal?